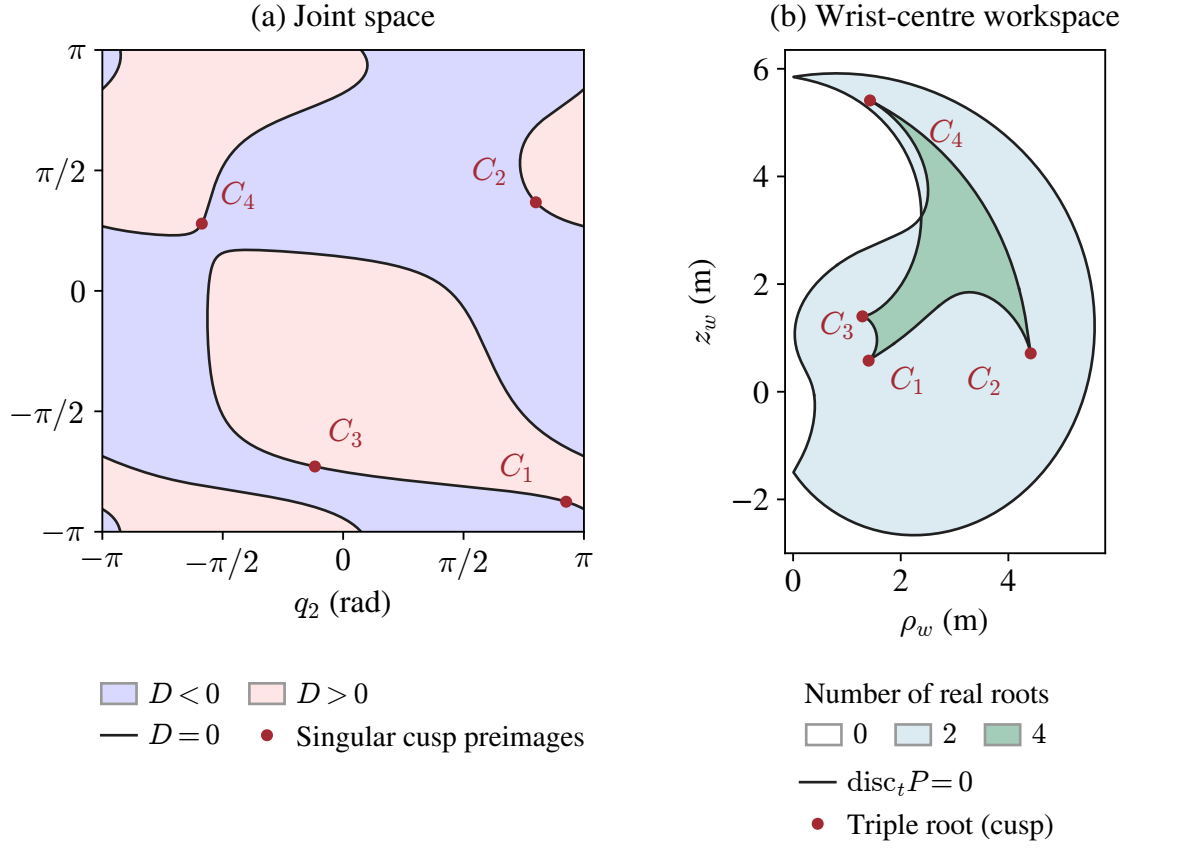

(a) Joint space
(b) Wrist-centre workspace
π
$\pi/2$
0
$-\pi/2$
$-\pi$
C_1
C_2
C_3
C_4
q_2 (rad)
6
4
2
0
−2
z_w (m)
ρ_w (m)
$D<0$
$D>0$
$D=0$
Singular cusp preimages
Number of real roots
0
2
4
$\mathrm{disc}_t P=0$
Triple root (cusp)